

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1335

Galion Generating Station 500 kV
May 03, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1335. The Interconnection Customer proposes a 72.6 MW solar photovoltaic facility interconnecting at the Galion Generating Station 500 kV substation in the Ohio area. The request was received on November 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 434 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,387,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cannon Branch-Galion Generating Station 230kV	Myrtle Pt Substation-Galion Generating Sta 230kV	153.3	565	15.2
Mcmillan-Cannon Branch 230kV	Myrtle Pt-Galion Generating Sta 230kV	153.2	559	15.2
Waldo-Casto 230kV Line	Galion Generating Station 500/230 kV XFMR	137.4	517	11.3
Galion Generating Sta-Casto 230kV L	Galion Generating Sta 500 kV/230 kV XFMR	135.6	499	11.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Galion Generating Station-Cannon Branch 230kV	Rebuild line with higher-capacity conductor	\$19,513,000	58% (\$11,306,000)
Mcmillan-Cannon Branch 230kV	Rebuild line with higher-capacity conductor	\$25,045,000	58% (\$14,511,000)
Waldo-Casto 230kV	Rebuild line with higher-capacity conductor	\$17,748,000	47% (\$8,359,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Galion Generating Sta-Casto 230kV line	Rebuild line with higher-capacity conductor	\$47,159,000	47% (\$22,211,000)
	Total allocated to project		\$56,387,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.